3.3.4
Future of jobs: insights from industry leaders

A Emerging job roles in the sector

Based on the survey conducted with industry leaders in the infrastructure sector, several emerging job roles are anticipated within the next decade. These roles are primarily driven by advancements in technology, sustainability and the increasing complexity of infrastructure projects. The following key job roles are expected to see significant growth:

- **AI and Machine Learning specialists:** With the increasing adoption of AI and machine learning, there will be a high demand for specialists who can develop, implement, and manage these technologies within the infrastructure sector as per 67% of the respondents.
- **Renewable energy technicians:** 67% of the respondents believe that as the focus on sustainability intensifies, there will be a growing need for technicians skilled in the installation, maintenance and optimization of renewable energy systems.
- **Sustainable infrastructure engineers:** Engineers with expertise in sustainable design and green technologies will be crucial as the sector moves towards more environment friendly practices as per 33% of the respondents.
- **Smart City planners:** 33% of the respondents believe that the development of smart cities will create a demand for planners who can integrate IoT, AI, and other advanced technologies into urban infrastructure.

B Impact of GenAI, AI, sector-specific technology, international mobility, DPI, climate, and green jobs on the job and skilling ecosystem

The infrastructure sector is on the cusp of significant transformation due to several technological and environmental factors:

- **AI and GenAI:** AI is expected to have a moderate impact on the infrastructure sector over the next two years, with key areas including autonomous construction vehicles, traffic management systems, and predictive maintenance. However, challenges such as the lack of technical skills (100% of respondents' observation) and high costs of implementation (33% of respondents' observation) may slow adoption.
- **Sector-specific technology:** Technologies like cloud computing and cybersecurity are already in use by 67% of respondents, and there is a growing interest in integrating more advanced tools like AI, 3D printing, and smart sensors. The sector is still lagging in the adoption of IoT and robotics, which presents opportunities for future growth.
- **Climate and green jobs:** The sector's move towards sustainability is evident, with 67% of respondents using renewable energy sources and eco-friendly materials in construction. However, the creation and filling of green jobs face challenges such as the lack of qualified candidates and resistance to changing traditional practices, as 1/3rd of the respondents observing this.
- **International mobility:** As infrastructure projects become more global, skills related to international trade and logistics are becoming increasingly important, with 67% of respondents recognizing the need for these competencies.

C Suggested programmatic and policy interventions

To meet the evolving needs of the infrastructure sector, both industry leaders and the government need to take the following actions:

Industry initiatives

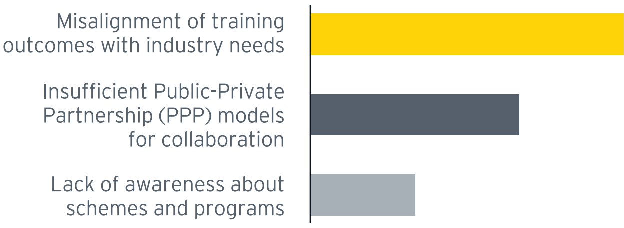
- **Reskilling and upskilling:** With the anticipated impact of AI and other technologies, 67% of the respondents agree reskilling and upskilling programs to be essential to prepare the workforce for new roles and responsibilities
- **Sustainability practices:** 67% of the respondents confirmed that their organization has adopted sustainable practices such as use of renewable energy sources (solar, wind, hydro) and eco-friendly materials in construction followed by water recycling and implementation of green building standards

Government Intervention

- **Financial incentives:** 100% of respondents identified the need for financial incentives for green investments. The government should introduce subsidies and tax credits to encourage the adoption of sustainable practices

- Support for technology adoption: While 67% of respondents find current policies somewhat effective, there is a clear demand for more targeted support, especially in providing grants for new technology and training employees
- 100% of the respondents believed that the skilling institutions should focus on sector specific technology interventions

Challenges infrastructure companies face in collaborating with educational and skilling institutions



The future of jobs in the infrastructure sector is heavily influenced by advancements in AI, sustainability and global mobility. There is a clear need for strategic interventions from both the industry and the government to address skill gaps, promote technology adoption, and ensure sustainable development. The anticipated job roles and required skills highlight the evolving landscape of the sector, where technology and sustainability will play pivotal roles in shaping the workforce of tomorrow.

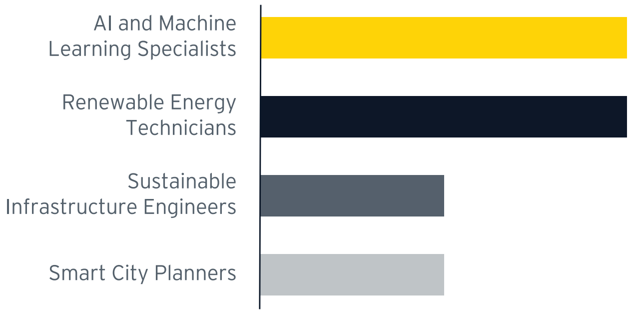
3.3.5
Overall impact on job roles and skills in the sector

As the infrastructure sector continues to integrate new technologies and sustainable practices, job roles are undergoing significant evolution. There is a growing demand for positions cantered on project management, data analysis, and environmental compliance, reflecting the sector's shift towards modernization and sustainability.

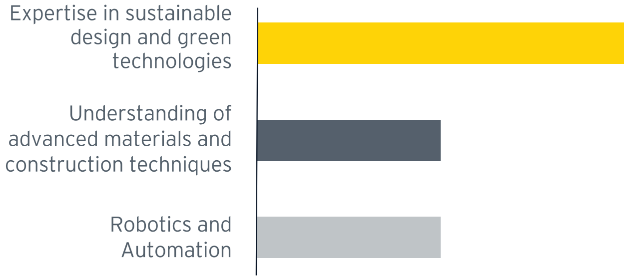
New roles, such as data centre engineers and site reliability engineers (SREs), are emerging, requiring specialized expertise in cloud technologies and automation. While the rise of automation and artificial intelligence (AI) may lead to a reduction in jobs involving routine and repetitive tasks, such as manual server provisioning and configuration management, it also opens the door to new opportunities. The demand for skills in AI, automation, and sustainable practices will increase, highlighting the need for continuous learning and adaptation within the workforce.

As automation streamlines manual tasks, the emphasis will shift towards skills that are uniquely human and difficult for machines to replicate, such as problem-solving, critical thinking, and adaptability. Despite the challenges posed by technological advancements, the ongoing investment in infrastructure and the transition to sustainable practices are expected to generate long-term employment opportunities, particularly for those with the right skills and qualifications.

Emerging job roles in infrastructure sector



Skill sets that will be most in demand for future infrastructure job roles



This evolving landscape underscores the critical importance of preparing the workforce to meet the future demands of the infrastructure sector, ensuring that they are well-equipped to thrive in a rapidly changing environment.

3.4 Energy

The energy sector in India is a dynamic and rapidly evolving segment of the economy, playing a crucial role in supporting the country's development and growth aspirations. It is characterized by a diverse mix of coal, oil, natural gas, renewable energy sources and nuclear power, along with energy efficiency and management services, and energy storage and battery solutions. The government has been actively promoting the adoption of renewable energy through ambitious targets and policy support, leading to an increase in solar and wind energy installations. The energy sector is also undergoing reforms aimed at improving efficiency and reducing carbon emissions. With a large and growing population, the demand for energy in India is expected to continue to rise, presenting both challenges and opportunities for the sector's expansion and modernization in emerging technologies.

3.4.1 Macroeconomic trends

A Key statistics

India is the third-largest producer and consumer of electricity globally, boasting an installed power capacity of 442.85 GW as of April 30, 2024. The country's growing population, coupled with increasing electrification and rising per-capita usage, will continue to drive this upward trajectory.

> In 2023, power consumption in India recorded a robust 9.5% growth, reaching 1,503.65 billion units (BU)[56].

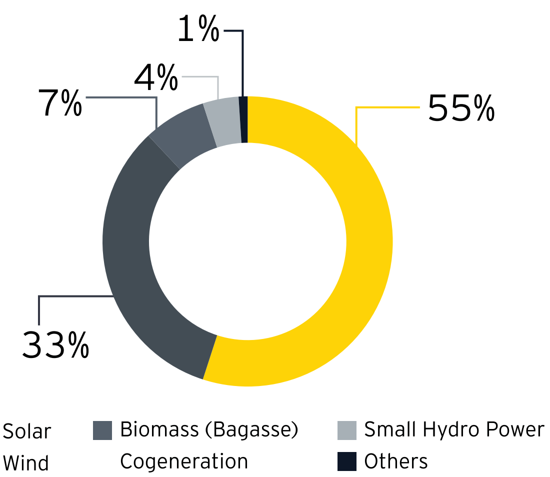
The energy sector is a cornerstone of India's economy, playing a pivotal role in both GDP and employment.

India is the third largest energy consuming country in the world[57]

- It has attracted substantial foreign direct investment (FDI), with total inflows amounting to $18.28 billion between April 2000 and March 2024, representing 2.69% of India's total FDI inflow
- Solar generation in India increased significantly from 7.45 BU in 2015-16 to about 102 BU in 2022-23 at a CAGR of 45.3%[58]
- Gross electricity generation in India has increased from 747.07 BU in 2008-09 to 1624.47 BU in 2022-23, at a CAGR of about 5.7%
- India's installed renewable energy capacity is expected to increase from 135 GW in December 2023 to about 170 GW by March 2025

India is making significant strides in the global shift towards renewable energy, with a commitment to achieving net-zero carbon emissions by 2070. India plans to add a considerable amount of renewable energy capacity annually over the next five years, a strategy that is expected to generate millions of new jobs within the sector[59]. This ambitious growth trajectory highlights India's leadership in advancing renewable energy and underscores the potential for substantial employment opportunities.

India Renewable Energy Mix (2024)



B Employment scenario

Employment in the energy sector is set to expand significantly. By 2030, global job numbers in the industry are projected to reach 139 million, with renewable energy alone contributing 38.2 million jobs and other energy transition technologies accounting for 74.2 million[60].

> In India, the sector employed nearly 1 million people in 2022, with a notable portion engaged in renewable energy projects. Solar PV, the fastest-growing segment, employed 2.82 lakh people in both on-grid and off-grid systems, with grid-connected solar PV jobs increasing by an impressive 47% to 2.01 lakh in 2022.

Hydropower continues to be the largest employer within the sector, with 4.66 lakh jobs. The wind sector also plays a vital role, providing 40,000 jobs, nearly half of which are in operations and maintenance. Overall, the solar and wind sectors employed 1.64 lakh workers in FY2022, reflecting a remarkable 47% increase from FY2021. This growth is largely driven by India's expanding renewable capacity, particularly in solar, which saw a record addition of 13.5 GW in 2022.

The government's ambitious target of achieving 500 GW of non-fossil fuel capacity by 2030 is projected to generate over 2 million jobs in the renewable energy sector.

C Key drivers of growth and government schemes

The energy sector in India is experiencing remarkable growth, fueled by government policies, technological advancements and the rising demand for sustainable energy. In recognition of the sector's importance, the government has allocated INR19,100 crore to the

Ministry of New and Renewable Energy in the Union Budget 2024-25, with a significant INR10,000 crore earmarked specifically for solar energy projects[61]. This reflects a 110% increase in budgetary allocation for solar initiatives compared to the previous year, underscoring the government's commitment to energy security and the promotion of innovative solutions such as pumped storage projects for electricity[62].

Key government initiatives are:

- **The National Infrastructure Pipeline (NIP)** has projected a substantial infrastructure investment of INR111 lakh crore during 2020-2025, with approximately 24% dedicated to the energy sector. Within this framework, an investment of INR9.295 lakh crore is projected for renewable energy, aiming to achieve an installed capacity of 265.73 GW by December 2025
- **The National Green Hydrogen Mission (NGHM)** is anticipated to create approximately 6 lakh jobs across the green hydrogen value chain by 2030[63], spanning electrolyzer and component manufacturing as well as green hydrogen production, highlighting the sector's commitment to sustainable energy solutions
- **The Deen Dayal Upadhyaya Gram Jyoti Yojana (DDUGJY)** and the Integrated Power Development Scheme (IPDS), are driving progress along with strong emphasis on solar and wind energy
- Additional government initiatives, such as the **Production Linked Incentive (PLI)** scheme for high-efficiency solar PV modules and the National Programme on High-Efficiency Solar PV Modules, are expected to enhance domestic manufacturing capabilities and generate numerous job opportunities in the solar energy sector underlining its potential for significant job creation and its transformative impact on the energy landscape

These initiatives highlight the sector's dynamic growth and its critical role in shaping India's sustainable future.

3.4.2
International benchmark

Ranking of India against global indices

India is currently ranked 63rd on the Global Energy Transition Index (ETI), which assesses countries based on their readiness for energy transition, considering factors such as economic development, environmental sustainability and energy security. This ranking highlights India's significant strides in enhancing energy equity, security and sustainability, driven by a strong emphasis on renewable energy. As of 2023, India stands as the third-largest solar power generator globally, boasting an installed solar capacity exceeding 65 GW.

International best practices

International best practices for integrating AI with sustainability in the energy sector emphasize several critical areas for effective implementation and long-term impact. AI significantly enhances efficiency through smart grids that enable real-time energy management and optimization. Predictive maintenance powered by AI helps reduce downtime and operational costs. Demand response systems optimize energy usage, while advanced algorithms improve renewable energy forecasting and energy storage management. The adoption of green technologies and robust cybersecurity measures is also crucial. Global collaboration plays a vital role in sharing policy frameworks and technological advancements, fostering innovation and resilience in the energy sector[64]. Leading examples include Denmark and Germany, which excel in integrating AI with sustainability through AI-driven green finance strategies and consistent renewable energy policies[65].

3.4.3
Sectoral trends

A Emerging technologies in the sector

The future skills demand in the energy sector is anticipated to centre around renewable energy technologies, energy storage, and digital skills. Government programs including the Green Energy Corridor Projects and the Solar Energy Corporation of India (SECI), will play a vital role in addressing these needs through skilling institutions and apprenticeship programs.

Some of the emerging technologies in the sector are enumerated below-

Solar energy innovations: Solar photovoltaic systems, wind turbines, and other renewable energy technologies are becoming more efficient and cost-effective. Innovations like floating solar farms and offshore wind turbines are opening new possibilities for energy generation.

Smart grids and smart meters: The adoption of smart grids and smart meters can improve energy efficiency and manage demand. The integration of smart grids with IoT devices allows for real-time monitoring and management of energy flow. This leads to increased grid stability, optimized energy distribution, and the facilitation of distributed energy resources.

Energy storage solutions: The energy storage technologies include advanced battery systems and pumped hydro storage projects.

Electric mobility: The rise of EVs is driving the need for more efficient batteries and charging infrastructure. Vehicle-to-grid (V2G) technology also allows EVs to return energy to the grid during peak demand.

Digitalization and AI: The use of AI, machine learning, and data analytics is growing in energy sector for predictive maintenance, load forecasting and optimizing grid operations.

As the sector adopts new technologies and transitions towards more sustainable energy production and distribution, technical proficiency and critical thinking skills become increasingly essential.

B Workforce trends

The Indian energy sector is undergoing a dynamic transformation as it moves towards cleaner, more advanced energy solutions. This transition is not only creating new job opportunities but also facilitating workforce mobility from traditional roles to emerging green jobs in solar, wind and energy storage. Some of the workforce trends in the energy sector:

Increasing demand for renewable energy skills: As India aggressively pursues renewable energy targets, there is a growing need for professionals with expertise in solar, wind, hydro, and biomass energy technologies. This includes roles in project planning, installation, operation and maintenance.

Diversification of roles: The energy sector is no longer limited to conventional roles. It now encompasses a variety of positions in areas such as energy storage, smart grid technology, energy efficiency, and electric vehicle infrastructure.

Adoption of digital tools: As digital technologies become integral to energy operations, there is a higher demand for professionals skilled in data analytics, cybersecurity, and software development within the energy domain.

Regulatory and policy expertise: As the energy sector evolves, there is a need for professionals who understand the regulatory and policy aspects of energy, including international agreements.

The sector demonstrates significant potential for workforce growth, with 62% of employers planning to expand their workforce during the first half of FY 2024-25. This anticipated growth is driven by the sector's transition towards a low-carbon future, supported by increased capital expenditure, incentives for clean

energy, and private sector investment. Embracing these trends will not only support the sector's evolution but also contribute to its continued success and resilience.

3.4.4 Future of jobs: insights from industry leaders

A Emerging job roles in the sector

The energy sector is expected to witness the creation of several new job roles in the coming years. Key roles identified include:

- **Renewable Energy technicians:** 40% of respondents anticipate growth in this role, reflecting the sector's shift towards renewable energy sources.
- **Smart Grid analysts:** Also expected by 40% of respondents, this role will be crucial in managing the increasingly complex grid systems.
- **Sustainability managers:** With a focus on environmental impact, 40% of respondents foresee the need for roles dedicated to managing sustainability initiatives.
- **Carbon Capture technicians:** As carbon management becomes a priority, this role is expected to grow, indicated by 40% of respondents.

These roles will require specialized skills in renewable energy technologies, energy storage solutions, and energy data analytics.

B Impact of GenAI, AI, sector-specific technology, international mobility, DPI, climate, and green jobs on the job and skilling ecosystem

The energy sector is on the cusp of significant transformation due to several technological and environmental factors:

AI and GenAI: AI and GenAI are expected to have a profound impact on the Energy sector, particularly in areas like renewable energy forecasting, predictive maintenance, and energy consumption analytics. However, there are challenges such as the technical complexity of integration, high costs, workforce resistance, and a lack of technical skills, with 60% of respondents citing these concerns.

Sector-specific technology: Technologies like the Internet of Things (IoT), AI, and Machine Learning (ML) are already in use, with 67% and 50% of respondents utilizing these, respectively. However, the adoption of more advanced technologies like robotics, additive manufacturing, and cloud computing remains limited.

International mobility: Skills related to international trade, logistics, and cross-cultural competencies will become increasingly important. 60% of respondents believe that these skills will be crucial as the sector becomes more globally integrated.

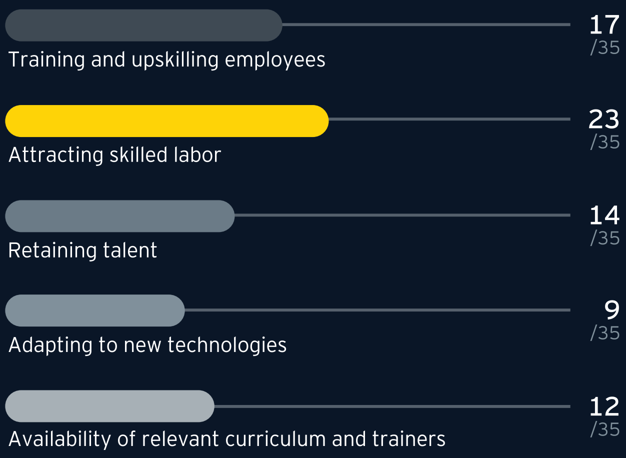
Utilization of any of the Industry 4.0 technologies

67%	Internet of Things (IoT)
50%	Artificial Intelligence (AI) and Machine Learning
17%	Robotics and Automation
50%	Big Data and Analytics
17%	Cloud Computing
33%	Cybersecurity measures

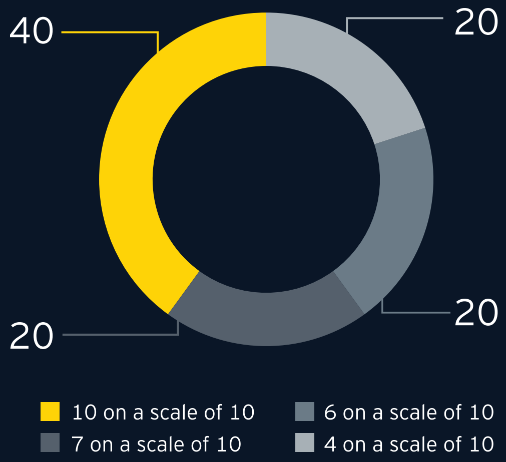
Sustainable practices organizations in the sector have adopted

40%	Renewable Energy Development
80%	Energy Efficiency Initiatives
60%	Use of electric vehicles (EVs)

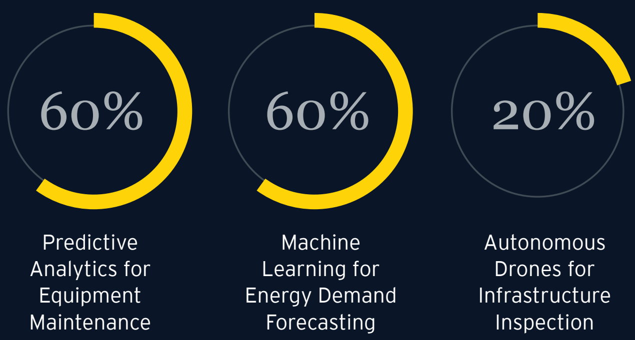
Biggest challenge in workforce development in Energy operations



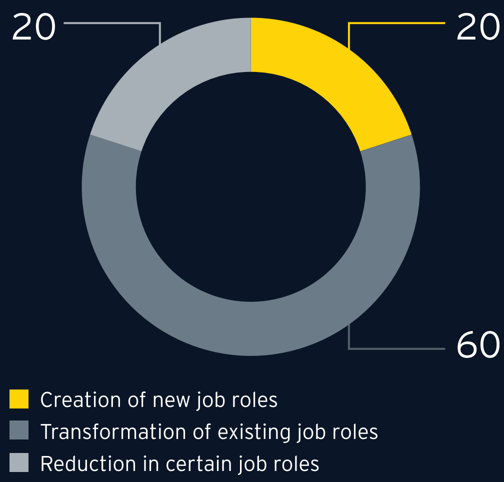
Significance of positive impact of AI and Gen AI on the energy sector in the next two years



AI applications are currently in use or planned for use in the sector

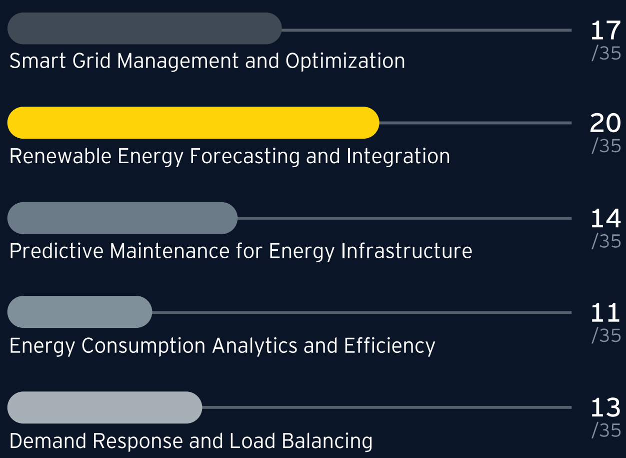


Impact of adoption of AI had on job roles



Anticipated challenges in integrating AI into existing energy systems

Areas of Energy sector where AI is expected to have the greatest impact



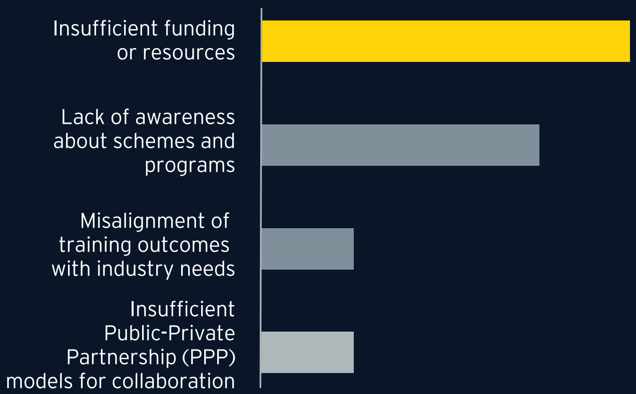
C Suggested programmatic and policy interventions

To meet the evolving needs of the energy sector, both industry leaders and the government need to take the following actions:

Industry initiatives

- **Upskilling initiatives:** With 60% of respondents identifying the need for reskilling programs, industry leaders should prioritize upskilling the workforce to adapt to AI and other technological advancements.
- **Sustainability practices:** While energy efficiency initiatives are common (80% adoption), there is a need to expand the adoption of practices like smart grid technology and sustainable bioenergy.

Challenges energy companies face in collaborating with educational and skilling institutions



Skills that will be crucial for future energy professionals for workforce mobility

40% Multilingual communication abilities

40% Cross-cultural competency and adaptability

60% International trade and logistics knowledge

40% Remote operation and telepresence skills

40% Global regulatory and compliance awareness

Skill sets to be most in demand for future energy job roles

60% Expertise in Renewable Energy Technologies

40% Proficiency in Energy Storage Solutions

60% Skills in Energy Data Analytics and AI

20% Knowledge of Environmental and Energy Policy

Government incentives or subsidies can be/have been most beneficial for technological advancements in the sector

60% Tax credits for research and development

80% Grants for purchasing new equipment or software

40% Subsidies for training employees in new technologies

80% Incentives for adopting sustainable practices

How well do you think current educational curricula align with the skill requirements of the Energy sector

40% Somewhat well

60% Neutral

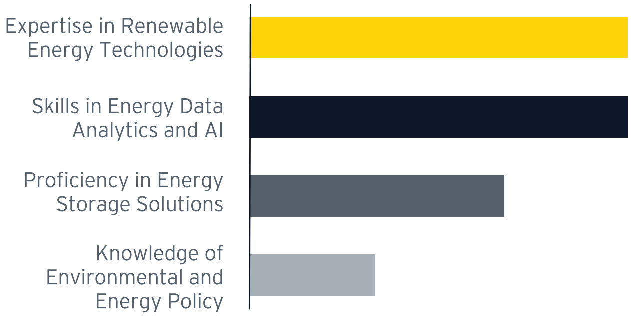
The future of jobs in the energy sector will be shaped by the rapid adoption of AI, the transition to sustainable practices, and the increasing importance of international mobility. To harness these opportunities, both industry leaders and the government must work together to address the challenges of workforce development, technological integration and sustainability.

3.4.5
Overall impact on job roles and skills in the sector

The energy sector in India is undergoing a significant transformation, driven by advancements in technology, a strong push towards sustainability, and evolving government policies. Primary and secondary research highlights a substantial shift in job roles towards renewable energy technologies, smart grid management, and sustainability-focused positions. Emerging roles such as Renewable Energy Technicians, Smart Grid Analysts, and Carbon Capture Technicians are gaining prominence, requiring specialized skills in AI, energy analytics, and environmental policy. This transition underscores the need for reskilling and upskilling programs to meet the sector's evolving demands.

Emerging job roles in energy sector

Skill sets that will be most in demand for future energy job roles



Areas skilling institutions need to focus on